Sowing Seasons

Planting calendars, frost dates, and crop timing for the United States

A detailed reference guide • United States
Compiled October 2026

Note: *This guide compiles general, widely accepted agronomic practice for US growers. Conditions vary by state, soil, and season — confirm specifics with your local Cooperative Extension office (find yours at nifa.usda.gov/land-grant-colleges-and-universities-partner-website-directory) and always follow current product labels and local regulations.*

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1. How the US Growing Season Works

In most of the United States, planting dates are governed by two things: temperature (both air and soil) and the length of the frost-free period between the last spring frost and the first fall frost. Cool-season crops tolerate light frost and are planted early; warm-season crops are killed by frost and must wait until the soil warms and danger of frost has passed.

Because the US spans from subtropical Florida and Texas to short-season northern tiers, there is no single national calendar. The practical approach is to find your last and first frost dates and your USDA hardiness zone, then count each crop's days-to-maturity backward and forward from those anchors.

Note: *The single most useful number you can know for your location is your average last spring frost date. Nearly every warm-season planting decision is made relative to it: "2 weeks before," "on the frost date," "2 weeks after."*

2. USDA Hardiness Zones

The USDA Plant Hardiness Zone Map divides the country into zones based on the average annual minimum winter temperature, in 10°F bands (each split into "a" and "b" halves of 5°F). Zones matter most for perennial survival (fruit trees, berries, perennial forages) but also indicate general season length and timing for annual crops.

Zone	Avg. annual min temp	Representative areas
3	-40 to -30°F	Northern MN, ND, northern ME
4	-30 to -20°F	Upper Midwest, northern New England, MT
5	-20 to -10°F	IA, southern WI, OH valley, parts of CO
6	-10 to 0°F	KS, MO, southern OH, PA, NJ
7	0 to 10°F	VA, TN, OK, northern TX, OR/WA valleys
8	10 to 20°F	GA, central TX, coastal Carolinas, AZ
9	20 to 30°F	FL (central), south TX, coastal CA, AZ
10–11	30 to 50°F	South FL, Gulf Coast tips, Hawaii

Note: *Mason, Ohio and most of southern Ohio fall in Zone 6 (roughly 6a–6b). The planting windows in this guide are built around a Zone 5–6 reference, with regional adjustments noted in Section 7.*

3. Frost Dates & Season Length

Frost dates are averages, not guarantees — in any given year frost may come earlier or later. They are usually given as a probability (e.g., the date after which there is only a 10% or 50% chance of a damaging frost). Conservative growers use the later, safer dates for tender crops.

Zone	Avg. last spring frost	Avg. first fall frost	Frost-free days
3	late May – early Jun	early Sep	~90–120
4	mid–late May	mid Sep	~110–135
5	late Apr – mid May	early–mid Oct	~150–170
6	mid Apr – early May	mid–late Oct	~170–190
7	early–mid Apr	late Oct – early Nov	~190–220
8	mid Mar – early Apr	mid Nov	~230–260
9+	Feb or none	Dec or none	280–365

To find exact local dates, enter your ZIP code at a frost-date lookup (NOAA/NWS data, or your state Extension service). Microclimates matter: low-lying "frost pockets," valleys, and areas away from water bodies frost earlier.

4. Cool-Season vs Warm-Season Crops

Sorting crops into cool- and warm-season groups is the key to timing. Cool-season crops germinate and grow in cool soil and tolerate frost; warm-season crops need warm soil and are damaged or killed by frost.

Group	Soil temp to plant	Frost tolerance	Examples
Cool-season (hardy)	40–50°F	Tolerates hard frost	Peas, spinach, onions, kale, broccoli, lettuce, radish, cabbage, carrots
Cool-season (half-hardy)	45–55°F	Tolerates light frost	Beets, chard, cauliflower, potatoes
Warm-season (tender)	60–65°F	Killed by frost	Beans, corn, tomato, cucumber, squash, melon
Warm-season (very tender)	65–70°F+	Needs warm soil & nights	Peppers, eggplant, okra, sweet potato, cotton, melons

5. Field-Crop Planting Windows

Typical windows for major US field crops. These shift roughly two weeks later for each zone colder (north) and two weeks earlier for each zone warmer (south). Always plant into adequately warm, workable soil rather than strictly by the calendar.

Crop	Typical US planting window	Notes
Corn (field)	late Apr – late May (Corn Belt)	Soil ≥ 50°F; earlier in South (Mar), later far North
Soybean	early May – mid Jun	Soil ≥ 55°F; can follow small grains (double-crop) in South

Crop	Typical US planting window	Notes
Winter wheat	Sep – Oct (sown in fall)	After "Hessian fly-free" date; harvest next Jun–Jul
Spring wheat	Apr – May	Northern Plains; as soon as soil is workable
Cotton	mid Apr – May	Soil ≥ 60–65°F; Deep South and Southwest only
Grain sorghum	May – Jun	Warm soil; drought-tolerant
Rice	Apr – May	Gulf states, AR, CA; flooded fields
Alfalfa	spring (Apr–May) or late summer (Aug)	Perennial; avoid midsummer heat for establishment

6. Vegetable Planting Calendar

Timing is expressed relative to your last spring frost (LSF) and first fall frost (FFF). "Direct" means seed straight into the garden; "transplant" means start indoors first and set out hardened seedlings.

Spring plantings

Crop	When to plant	Method
Peas, spinach	4–6 wks before LSF	Direct
Onions, leeks	4–6 wks before LSF	Transplant/sets
Lettuce, radish, kale	3–5 wks before LSF	Direct
Broccoli, cabbage, cauliflower	2–4 wks before LSF	Transplant
Carrots, beets, chard	2–3 wks before LSF	Direct
Potatoes	2–4 wks before LSF	Seed pieces
Beans, corn	on–to 1 wk after LSF	Direct
Tomatoes	1–2 wks after LSF	Transplant
Peppers, eggplant	2–3 wks after LSF	Transplant
Squash, cucumber, melon	1–2 wks after LSF	Direct/transplant

Fall & overwinter plantings

Crop	When to plant	Notes
Fall brassicas (broccoli, cabbage)	10–12 wks before FFF	Mature in cool fall weather
Fall lettuce, spinach, radish	4–8 wks before FFF	Cool-season; sweeter after light frost
Garlic	3–6 wks before ground freezes (Oct–Nov)	Overwinters; harvest next summer

Crop	When to plant	Notes
Cover crop (cereal rye)	Sep – Oct	Overwinters, terminate in spring

7. Regional Timing Notes

Midwest / Corn Belt (Zones 4–6)

Core corn-and-soybean country. Corn planting concentrates late April into mid-May; soybeans follow through early June. Short enough season that full-season hybrids must go in on time. Fall is the window for winter wheat and cover crops.

Northeast & Upper Midwest (Zones 3–5)

Shorter seasons; start warm-season crops indoors and transplant. Last frost can linger into late May. Choose early-maturing varieties. Fall frost arrives by early–mid October.

South & Southeast (Zones 7–9)

Long seasons allow two or even three crops a year (double-cropping). Cool-season vegetables are often grown in fall through spring; summers can be too hot for them. Cotton, peanuts, and warm-season crops thrive.

Great Plains (Zones 4–7)

Winter wheat dominates the southern/central plains (sown fall, harvested early summer); spring wheat in the north. Moisture, not just temperature, often sets planting timing.

West & Pacific (Zones vary widely)

Highly variable by elevation and irrigation. Mediterranean California allows nearly year-round vegetable production; interior and mountain West have short, frost-bounded seasons. Irrigation scheduling is central.

8. Succession & Relay Planting

To keep a continuous supply and make full use of the season, stagger plantings rather than sowing everything at once.

- **Succession sowing.** Plant small amounts of fast crops (lettuce, radish, beans, sweet corn) every 1–3 weeks for a steady harvest instead of a glut.
- **Relay / double-cropping.** Follow an early crop with a second in the same season (e.g., soybeans after winter wheat in the South; fall greens after spring peas).
- **Intercropping.** Pair a fast crop with a slow one (radish between carrots) to use space before the slow crop fills in.
- Track days-to-maturity so each successive planting still has time to finish before frost.

9. Reading Soil Temperature

Soil temperature, not the calendar, ultimately determines germination. Seed sown into cold soil sits, rots, or emerges weakly. Measure with a soil thermometer at seeding depth, mid-morning, for several days to get a reliable average.

Crop	Minimum soil temp	Ideal soil temp
Peas, spinach, lettuce	40°F	50–65°F
Onions, brassicas	45°F	55–75°F
Corn	50°F	60–85°F
Beans	55–60°F	70–80°F
Tomato, cucumber, squash	60°F	70–90°F
Peppers, eggplant, okra, cotton	65°F	75–90°F

10. Planning Backward from First Frost

For fall and long-season crops, work backward from your average first fall frost to make sure a crop can finish in time:

- Start with the crop's days-to-maturity (on the seed packet or catalog).
- Add a “short-days factor” of about 2 weeks for fall crops, because growth slows as days shorten and cool.
- For transplants, count from set-out date; for direct seed, from germination.
- Count that total back from the first frost date to find the latest safe planting date.

Example: a 70-day broccoli crop for fall harvest, with a mid-October first frost, should be transplanted by roughly early August (70 days + ~2 weeks buffer before frost).

Note: *Row covers, low tunnels, and cold frames can extend both ends of the season by several weeks, letting you plant earlier in spring and harvest later into fall.*